

Shor's Factoring Algorithm and Public-Key Cryptography

Session 6 – Day 2

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Learning Objectives

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- Trace step-by-step factoring examples for $N = 15$ and $N = 21$.
- Learn how classical post-processing via continued fractions yields the period r .

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 - Public key exponent e satisfies $\gcd(e, \varphi(N)) = 1$, where $\varphi(N) = (p - 1)(q - 1)$.
 - Private key exponent d satisfies $ed \equiv 1 \pmod{\varphi(N)}$.
- **The Factoring Threat:** If an adversary can factor N to find p and q , they can compute $\varphi(N)$ and reconstruct the private key d from the public exponent e .

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- As long as $a^{r/2} \not\equiv -1 \pmod{N}$, the composite number N must share a non-trivial factor with at least one of $(a^{r/2} - 1)$ or $(a^{r/2} + 1)$.
- We can efficiently find these factors by computing:

$$\gcd(a^{r/2} - 1, N) \quad \text{and} \quad \gcd(a^{r/2} + 1, N)$$

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❹ **Compute GCDs:**

$$\gcd(2^2 - 1, 15) = \gcd(3, 15) = 3$$

$$\gcd(2^2 + 1, 15) = \gcd(5, 15) = 5$$

We successfully recover the prime factors 3 and 5.

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We successfully recover the prime factors 7 and 3.

Order Finding as Eigenphase Estimation

- Define the modular multiplication unitary operator U_a by:

$$U_a |s\rangle = |as \pmod N\rangle$$

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- Finding the order r is equivalent to estimating these eigenphases $\phi = k/r$ using Quantum Phase Estimation (QPE).

Eigenstates of the Modular Unitary

- For each $k \in \{0, 1, \dots, r-1\}$, we define the eigenstate:

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- By preparing the target register in the trivial state $|1\rangle$, we are effectively running QPE on a uniform superposition of all eigenstates $|u_k\rangle$ simultaneously!

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2 Controlled Modular Exponentiation: Apply controlled powers of U_a :

$$|\psi_1\rangle = \frac{1}{\sqrt{Q}} \sum_{x=0}^{Q-1} |x\rangle |a^x \pmod N\rangle$$

This entangles the registers and embeds the periodic structure of modular multiplication.

Period Decoupling and Measurement

- Expanding the target state in terms of the eigenstates $|u_k\rangle$:

$$|\psi_1\rangle = \frac{1}{\sqrt{r}} \sum_{k=0}^{r-1} \left[\frac{1}{\sqrt{Q}} \sum_{x=0}^{Q-1} e^{2\pi i k x / r} |x\rangle \right] |u_k\rangle$$

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- Measuring the control register yields an integer x which, with high probability, satisfies:

$$\frac{x}{Q} \approx \frac{k}{r} \quad \text{for some } k \in \{0, 1, \dots, r-1\}$$

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- We verify this candidate by checking if $a^{q'} \equiv 1 \pmod{N}$. If not, we can test small multiples of q' or repeat the quantum subroutine to get a different k .

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- 5 If r is even, compute $a^{r/2} \pmod{N}$.
- 6 If $a^{r/2} \equiv -1 \pmod{N}$, declare failure and go back to step 1.
- 7 Otherwise, compute $\gcd(a^{r/2} - 1, N)$ and $\gcd(a^{r/2} + 1, N)$. At least one of these is a non-trivial factor of N .

Intuition: From Waves to Frequencies

The Waveform Paradigm

- ****Classically****, the modular sequence $a^1, a^2, a^3, \dots$ modulo N behaves like a hard, pseudorandom sequence. Finding the period requires computing terms step-by-step.

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- **Quantumly**, we load the entire sequence into a single superposition, converting it into a periodic wave of quantum amplitudes.

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- **Quantumly**, we load the entire sequence into a single superposition, converting it into a periodic wave of quantum amplitudes.
- The **Inverse QFT** acts like a frequency analyzer (spectrometer), taking the periodic wave and resolving it into a sharp frequency peak at $1/r$.

Intuition: From Waves to Frequencies

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- The **Inverse QFT** acts like a frequency analyzer (spectrometer), taking the periodic wave and resolving it into a sharp frequency peak at $1/r$.
- Measuring this peak yields the frequency, which we classically invert to find the period.

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Problem: Prove that the eigenstates $|u_k\rangle$ satisfy $U_a |u_k\rangle = e^{2\pi i k/r} |u_k\rangle$.

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- ④ Since $a^r \equiv 1 \pmod{N}$ and $e^{-2\pi i k r/r} = 1$, the term for $s' = r$ is identical to $s' = 0$. Thus the sum over s' is exactly $|u_k\rangle$:

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- Compute the ratio: $\frac{x}{Q} = \frac{64}{256} = \frac{1}{4}$.
- This is already in lowest terms. The candidate denominator is $r = 4$.

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Case (b): Measurement $x = 85$

- Compute the ratio: $\frac{x}{Q} = \frac{85}{256} \approx 0.33203$.
- Apply continued fractions expansion:

$$\frac{85}{256} = \frac{1}{\frac{256}{85}} = \frac{1}{3 + \frac{1}{85}} \approx \frac{1}{3}$$

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- 3 The core quantum machinery is Quantum Phase Estimation on the eigenvalues of the modular multiplication operator.
- 4 Periodicity is converted into peaks in measurement statistics using the Inverse Quantum Fourier Transform.
- 5 The vulnerability of public-key cryptosystems makes the transition to Post-Quantum Cryptography (PQC) urgent.